

Program : Diploma in Tool & Die Engineering	
Course Code : 6117	Course Title: CAM & 3D Printing Lab
Semester : 6	Credits: 2.5
Course Category: Program Elective	
Periods per week: 4 (L:1, T:0, P:3)	Periods per semester: 60

Course Objectives:

- To introduce the various features in the menu of solid modeling package.
- To explain the synthesis of various parts or components in an assembly.
- To familiarize the students with CNC programs for various jobs.
- To familiarize the students with the rapid prototyping through 3D printing.

Course Prerequisites:

Topic	Course code	Course name	Semester
CNC part programming basics and awareness of Rapid prototyping techniques		Modern production process	5

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the working of CNC turning and milling machine.	15	Applying
CO2	Prepare the part program using simulation software for Lathe and Milling.	14	Applying
CO3	Prepare the part program, edit and execute in CNC turning and machining Centre	16	Applying
CO4	Demonstrate the working of 3D printer and rapid prototyping	12	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			3
CO3	3			3			3
CO4	3			3			3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the working of CNC turning and milling machine		
M1.01	Demonstrate CNC Programming and Machining: Introduction: Study of CNC lathe, milling	2	Applying
M1.02	Interpret international standard codes: G-Codes and M-Codes	2	Applying
M1.03	Use Format – Dimensioning methods	2	Applying
M1.04	Prepare Program – Turning simulator – Milling simulator, IS practice – commands menus	3	Applying
M1.05	Modify the program in the CNC machines	3	Applying
M1.06	Apply the program in the CNC machines; Exercises: Note: Print the Program from the Simulation Software and make the Component in the CNC Machine.	3	Applying
CO2	Prepare the part program using simulation software for Lathe and Milling.		
M2.01	Produce component in CNC Turning Machine with part program (Linear and Circular interpolation, Material: Aluminum/Acrylic/Plastic rod)	4	Applying
M2.02	Produce component with multiple turning operations in CNC Turning Machine with Part program (Stock removal cycle)	5	Applying

M2.03	Produce component with thread and grooving in the CNC Turning Machine with Part program (canned Cycle)	5	Applying
	Lab Exam – I	1.5	
CO3	Prepare the part program, edit and execute in CNC turning and machining Centre.		
M3.01	Produce component with grooving in CNC Milling Machine with Part program (Linear interpolation and Circular interpolation, Material: Aluminum/ Acrylic/ Plastic	8	Applying
M3.02	Produce component with drilling, tapping and countersinking in the CNC Milling Machine. (Canned cycle)	4	Applying
M3.03	Produce component with mirroring in the CNC Milling Machine with Part program (sub program)	4	Applying
CO4	Demonstrate the working of 3D printer and rapid prototyping		
M4.01	Explain 3D printing	2	Understanding
M4.02	Demonstrate Prototype design	3	Applying
M4.03	Prepare 3D printing programs & tool path generation	4	Applying
M4.04	Practice Printing of objects with PLA/ABS materials	3	Applying
	Lab Exam – II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	CAD/CAM/CIM – P. Radhakrishnan, S. Subramaniyan& V. Raju, New Age International Pvt. Ltd., New Delhi, 3rd Edition,
R1	Manufacturing process for engineering materials – Serope Kalpakjian
R2	N.C. Machines and CAM - Pressman Williams

Online Resources:

Sl.No	Website Link
1	https://www.engineeringbookspdf.com/download/?file=4559
2	https://www.instructables.com/id/UNDERSTANDING-CNC-AND-BASIC-PART-PROGRAMMING-FOR-M/
3	https://3dprintingindustry.com/3d-printing-basics-free-beginners-guide#04-processes